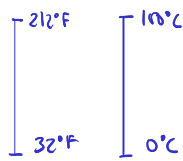


### Problema 1



$$^{\circ}\text{F} = 1,8 ^{\circ}\text{C} + 32^{\circ}\text{F}$$

$$X = 1,8 X + 32 \Rightarrow X(1-1,8) = 32 \quad X = \frac{32}{-0,8} = -40$$

$$-40^{\circ}\text{F} = -40^{\circ}\text{C}$$

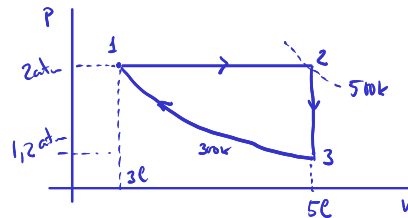
### Problema 2



$$\Delta U = Q - W = 800 \text{ cal} - 2000 \text{ J} = 800 \text{ cal} \cdot \frac{4,18 \text{ J}}{1 \text{ cal}} - 2000 \text{ J} =$$

$$= 1352 \text{ J}$$

### Problema 3



$$p_1 = 2 \text{ atm} \quad T_1 = 300 \text{ K} \quad V_1 = 3 \text{ L}$$

$$C_v = 3 \text{ cal/mol K} \quad R = \frac{2 \text{ cal}}{\text{mol K}} = 0,082 \frac{\text{atm L}}{\text{mol K}}$$

|   | atm | L  | K    |
|---|-----|----|------|
| P | V   | T  |      |
| 1 | 2   | 3  | 300  |
| 2 | 2   | 5  | 500  |
| 3 | 1,2 | 3L | 300K |

A) Del punto 1.

$$n = \frac{PV}{RT} = \frac{2 \text{ atm} \cdot 3 \text{ L}}{0,082 \frac{\text{atm L}}{\text{mol K}} \cdot 300 \text{ K}} = 0,2439 \text{ moles.}$$

Punto 2  $V_2 = \frac{nRT_2}{P_2} = \frac{0,2439 \text{ mol} \cdot 0,082 \frac{\text{atm L}}{\text{mol K}} \cdot 500 \text{ K}}{2 \text{ atm}} = 5 \text{ L}$

Punto 3  $P_3 = \frac{nRT_3}{V_3} = \frac{0,2439 \text{ mol} \cdot 0,082 \frac{\text{atm L}}{\text{mol K}} \cdot 300 \text{ K}}{3 \text{ L}} = 1,2 \text{ atm}$

B) isobárica  $p = \text{cte}$   $W_{12} = p \cdot \Delta V = 2 \text{ atm} \cdot (5 \text{ L} - 3 \text{ L}) = 4 \text{ atm} \cdot \text{L} \cdot \frac{2 \text{ cal}}{0,082 \text{ atm L}} = 97,6 \text{ cal}$

isocora  $V = \text{cte}$   $W_{23} = 0 \text{ cal}$

isoterma  $T = \text{cte}$   $W_{31} = nRT \ln(V_1/V_3) = p \cdot V_1 \cdot \ln(V_1/V_3) = 2 \text{ atm} \cdot 3 \text{ L} \cdot \ln(3/5) = -3,065 \text{ atm} \cdot \text{L} \cdot \frac{2 \text{ cal}}{0,082 \text{ atm L}} = -74,75 \text{ cal}$

$$W_{11} = W_{12} + W_{23} + W_{31} = 97,6 \text{ cal} - 74,75 \text{ cal} = 22,85 \text{ cal} = 95,7 \text{ J}$$

C) isobárica  $\Delta U_{12} = nC_v \Delta T = nC_v (T_2 - T_1) = nC_v (500 \text{ K} - 300 \text{ K}) = nC_v \cdot 200 \text{ K} = 613,16 \text{ J}$   
 isocora  $\Delta U_{23} = nC_v \Delta T = nC_v (T_3 - T_2) = nC_v (300 \text{ K} - 500 \text{ K}) = -nC_v \cdot 200 \text{ K} = -613,16 \text{ J}$   
 isoterma  $\Delta U = 0$

$$\Delta U_{11} = 0 \text{ J}$$

$$nC_v \cdot 200 \text{ K} = 0,2439 \text{ mol} \cdot \frac{3 \text{ cal}}{\text{mol K}} \cdot 200 \text{ K} = 146,34 \text{ cal} = 613,16 \text{ J}$$

$$C_p - C_v = R \Rightarrow C_p = R + C_v = 2 \text{ cal/mol} + 3 \text{ cal/mol} = 5 \text{ cal/mol}$$

D) isobárica  $Q_{12} = n C_p \Delta T = 0,2439 \text{ mol} \cdot 5 \text{ cal/mol} \cdot (500\text{K} - 300\text{K}) = 243,9 \text{ cal}$

isocora  $Q_{23} = n C_v \Delta T = 0,2439 \text{ mol} \cdot 3 \text{ cal/mol} \cdot (300 - 500\text{K}) = -146,34 \text{ cal}$

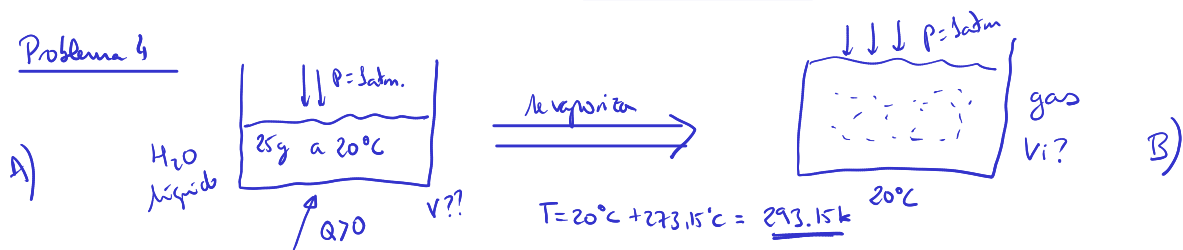
isoterma  $Q_{31} = W_{31} (\Delta U_{31} = 0) \quad Q_{31} = -74,75 \text{ cal}$

$$Q_{11} = Q_{12} + Q_{23} + Q_{31} = 22,81 \text{ cal}$$

$$W_{11} = 22,85 \text{ cal}$$

Como en el ciclo total  $\Delta U = 0 \Rightarrow Q = W$  y efectivamente; mismo decimal  $Q_{11} = W_{11} = 22,81 \text{ cal} \approx 95,6 \text{ J}$

#### Problema 4



$n = \frac{25\text{g}}{18\text{g/mol}} = 1,389 \text{ moles}$ .  $L = 580 \text{ cal/g}$  calor de vaporización

$Q = m \cdot L = 25\text{g} \cdot 580 \text{ cal/g} = 14500 \text{ cal}$

Calor que absorbe para cambiar de estado

$\rho = \frac{m}{V} \Rightarrow V = \frac{m}{\rho} = \frac{25\text{g}}{1\text{kg/l}} = 0,025 \text{ l}$

Al convertirse en calor se expande de  $V = 0,025 \text{ l}$  a... ?? Ahora es gas, luego.

$V = \frac{nRT}{P} = \frac{1,389 \text{ mol} \cdot 0,082 \frac{\text{atm} \cdot \text{l}}{\text{mol} \cdot \text{K}} \cdot 293,15 \text{ K}}{1 \text{ atm}} = 33,386 \text{ l}$

El trabajo que hace el gas es ( $p = \text{cte}$ ).

$W = p \cdot \Delta V = 1 \text{ atm} \cdot (33,386 \text{ l} - 0,025 \text{ l}) = 33,361 \text{ atm} \cdot \text{l} = 813,76 \text{ cal}$

$\Delta U = Q - W = 14500 \text{ cal} - 813,76 \text{ cal} = 13686,24 \text{ cal}$

#### Problema 5

$Q = m \cdot c \cdot \Delta T = \rho \cdot V \cdot c \cdot \Delta T = \frac{1 \text{ kg}}{\text{dm}^3} \cdot 5 \text{ dm}^3 \cdot \frac{1 \text{ cal}}{\text{g} \cdot ^\circ\text{C}} \cdot 25^\circ\text{C} \cdot \frac{1000 \text{ g}}{1 \text{ kg}} =$

$\Delta T = 25^\circ\text{C} = 25 \text{ K}$

$\rho = \frac{m}{V} \quad m = \rho \cdot V$

$= 125000 \text{ cal} = 125 \text{ kcal}$

¡Ojo! Agua líquida.

### Problema 6

$$Q = m \cdot c \cdot \Delta T$$

$$Q_{\text{metal}} = m \cdot c_{\text{metal}} \cdot \Delta T_{\text{metal}}$$

$$Q_{\text{agua}} = m \cdot c_{\text{agua}} \cdot \Delta T_{\text{agua}}$$

$$\frac{Q_{\text{metal}}}{Q_{\text{agua}}} =$$

$$\frac{c_{\text{metal}}}{c_{\text{agua}}} \cdot \frac{\Delta T_{\text{metal}}}{\Delta T_{\text{agua}}} = 1$$

$$\frac{\Delta T_{\text{metal}}}{\Delta T_{\text{agua}}} = \frac{c_{\text{agua}}}{c_{\text{metal}}} > 1$$

↓ 1, misma cantidad, 1000 cal.

$$\Delta T_{\text{metal}} > \Delta T_{\text{agua}}$$

### Problema 7



$$c_{\text{agua}} = \frac{Q_{\text{agua}}}{m \cdot \Delta T} = \frac{16 \text{ kcal}}{320 \text{ g} \cdot 50^\circ\text{C}} = 1 \text{ cal/g}^\circ\text{C}$$

$$c_{\text{glic.}} = \frac{Q_{\text{glic.}}}{m \cdot \Delta T} = \frac{9,28 \text{ kcal}}{320 \text{ g} \cdot 50^\circ\text{C}} = 0,58 \text{ cal/g}^\circ\text{C}$$

$$c_{\text{clor.}} = \frac{Q_{\text{clor.}}}{m \cdot \Delta T} = \frac{3,74 \text{ kcal}}{320 \text{ g} \cdot 50^\circ\text{C}} = 0,234 \text{ cal/g}^\circ\text{C}$$

### Problema 8

$$Q = m \cdot c \cdot \Delta T$$

$$m = 250 \text{ g}$$

$$c = 0,2 \text{ cal/g}^\circ\text{C}$$

$$T_i = 5^\circ\text{C} \quad T_F = 59^\circ\text{F} = 15^\circ\text{C}$$

$$F = 1,8 C + 32 \Rightarrow C = \frac{F - 32}{1,8} = \frac{59^\circ\text{F} - 32^\circ\text{F}}{1,8^\circ\text{F}/^\circ\text{C}} = \underline{15^\circ\text{C}}$$

$$Q = 250 \text{ g} \cdot 0,2 \frac{\text{cal}}{\text{g}^\circ\text{C}} \cdot (15^\circ\text{C} - 5^\circ\text{C}) = 500 \text{ cal}$$